

1) Free electrons:

$$H_e = \sum_k \epsilon_k a_k^\dagger a_k, \quad \epsilon_k = \frac{\hbar^2 k^2}{2m}$$

$$H_p = \sum_{\lambda} \omega_{\lambda} \left(b_{\lambda}^\dagger b_{\lambda} + \frac{1}{2} \right)$$

$$N_{\lambda} = \langle b_{\lambda}^\dagger b_{\lambda} \rangle = \frac{1}{e^{\omega_{\lambda}/T} - 1}$$

$$H_{e-p} = \sum_{\lambda, \sigma} \int d\vec{r} \psi_\sigma^\dagger(\vec{r}) \underbrace{U(\vec{r} - \vec{R}_\lambda)}_{U(\vec{r} - \vec{R}_\lambda^0) - \vec{u}_\lambda \cdot \vec{\nabla} U(\vec{r} - \vec{R}_\lambda^0)} \psi_\sigma(\vec{r})$$

$$H_{e-p} = - \sum_{\lambda, \sigma} \int d\vec{r} \psi_\sigma^\dagger(\vec{r}) \left(\vec{u}_\lambda \cdot \vec{\nabla} U(\vec{r} - \vec{R}_\lambda^0) \right) \psi_\sigma(\vec{r})$$

$$\vec{u}_\lambda = \frac{1}{N_i} \sum_{\vec{z}, \lambda} \sqrt{\frac{1}{2M\omega_{\lambda}}} \vec{z}_{\lambda} (b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^\dagger) e^{i\vec{z} \cdot \vec{R}_\lambda^0}$$

$$U = -\frac{ze^2}{r} e^{-\Lambda r}$$

$$\sum_{\vec{z}} e^{i(\vec{k} - \vec{k}' + \vec{z}) \cdot \vec{R}_\lambda^0} = N_i \sum_{\vec{G}} \delta_{\vec{k}, \vec{k}' + \vec{z} + \vec{G}}$$

$$- \sum_{\sigma, \vec{z}, \lambda} \frac{1}{\sqrt{2M\omega_{\lambda} N_i}} \vec{z}_{\lambda} \frac{c_{\vec{k}, \sigma}^\dagger c_{\vec{k}', \sigma}}{V} (b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^\dagger) e^{i\vec{z} \cdot \vec{R}_\lambda^0} \int d\vec{r} e^{i(\vec{k} - \vec{k}') \cdot \vec{r}} \underbrace{\vec{\nabla} \left(\frac{1}{|\vec{r} - \vec{R}_\lambda^0|} e^{-\Lambda |\vec{r} - \vec{R}_\lambda^0|} \right)}_{\substack{\vec{r} - \vec{R}_\lambda^0 = \vec{x} \\ \Lambda \vec{r} = -\vec{x}}}$$

$$\leftarrow \sum_{\vec{z}}$$

$$N_i = \frac{V}{\Omega}$$

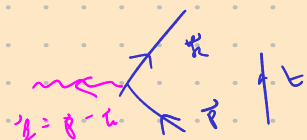
$$\sum_{\sigma, \vec{z}, \lambda} \frac{i \left[\vec{z}_{\lambda} \cdot (\vec{k} - \vec{k}') \right]}{\sqrt{2M\omega_{\lambda}}} \frac{1}{V} \sqrt{N_i} U(\vec{k} - \vec{k}') \sum_{\vec{G}} \delta_{\vec{k}, \vec{k}' + \vec{z} + \vec{G}} c_{\vec{k}, \sigma}^\dagger c_{\vec{k}', \sigma} (b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^\dagger) =$$

$$= \frac{1}{\sqrt{V}} \sum_{\vec{z}, \lambda, \sigma} g_{\lambda}(\vec{k}, \vec{z}) c_{\vec{k} + \vec{z}, \sigma}^\dagger c_{\vec{k}, \sigma} (b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^\dagger) ; \quad g_{\lambda}(\vec{k}, \vec{z}) = -i \frac{1}{\sqrt{2M\omega_{\lambda}}} \frac{4\pi ze^2}{(|\vec{z}|^2 + \Lambda^2)} \vec{z} \cdot \vec{z}_{\lambda}^*$$

→ Fermi: spin ne drinja!

$$|\langle f | c_{\vec{k}}^\dagger c_{\vec{r}} | i \rangle|^2 = n_{\vec{k}} (1 - n_{\vec{r}})$$

polu prazno



→ from trace: $K N_z - 1 |b_z| N_z \rangle|^2 = N_z$
 $| \langle N_z | b_z^\dagger | N_z + 1 \rangle |^2 = N_z + 1$

Fermi (variance): $\gamma_{i \rightarrow f} = 2\pi |K f | H_{fi} | i \rangle|^2 \underbrace{p(E_f)}_{\delta(E_f - E_i)}$

(1) Absorption process:

$|i\rangle = |N_z\rangle \otimes |k\rangle$
 $|f\rangle = |N_z - 1\rangle \otimes |k + z\rangle$
 $\rightarrow \sum_{k_z} \underbrace{\langle N_z - 1 | (b_z + b_{-z}^\dagger) | N_z \rangle}_{N_z} \underbrace{\langle k + z | c_{k+z}^\dagger c_k | k \rangle}_{\sqrt{n_k (1 - n_{k+z})}} \cdot g \frac{1}{\sqrt{V}}$

(2) emission process:

$|i\rangle = |N_z\rangle |k\rangle$
 $|f\rangle = |N_z + 1\rangle |k - z\rangle$
 $\rightarrow \sum_{k_z} \underbrace{\langle N_z + 1 | b_z^\dagger | N_z \rangle}_{\sqrt{N_z + 1}} \underbrace{\langle k - z | c_{k-z}^\dagger c_k | k \rangle}_{\sqrt{n_k (1 - n_{k-z})}} \cdot g \frac{1}{\sqrt{V}}$

(Mixture: $c_{k+z}^\dagger c_k b_z + c_{k-z}^\dagger c_k b_z^\dagger$)

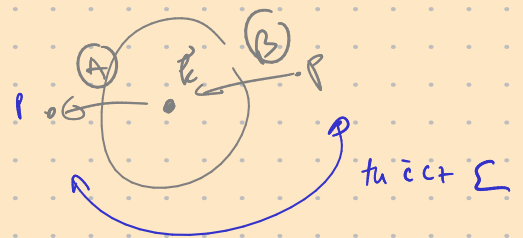
(A) $k \rightarrow p$: $\gamma_{k \rightarrow p} = \gamma_{abs} + \gamma_{em} = \frac{2\pi}{V} |g_\lambda(\vec{k}, \vec{z})|^2 n_k (1 - n_{k+z}) N_z \delta(\epsilon_{k+z} - \epsilon_k - \omega_z) +$
 $\frac{2\pi}{V} |g_\lambda(\vec{k}, \vec{z})|^2 n_k (1 - n_{k-z}) (N_z + 1) \delta(\epsilon_{k-z} + \omega_z - \epsilon_k)$

(B) $p \rightarrow k$: $\gamma_{p \rightarrow k} = \gamma_{abs} + \gamma_{em} = \frac{2\pi}{V} |g_\lambda(\vec{p}, \vec{z})|^2 n_{k+z} (1 - n_k) (N_z + 1) \delta(\epsilon_k + \omega_z - \epsilon_{k+z}) +$
 $\frac{2\pi}{V} |g_\lambda(\vec{p}, \vec{z})|^2 n_{k-z} (1 - n_k) N_z \delta(\epsilon_k - \epsilon_{k-z} - \omega_z)$

$N_{-z} = N_z$ then $\omega_z = \omega_{-z}$?

power in. strength

$I_k = \sum_p \gamma_{p \rightarrow k} - \gamma_{k \rightarrow p} = \dots$



$= \frac{2\pi}{V} \sum_{z, \lambda} |g_\lambda(\vec{k}, \vec{z})|^2 \left\{ (1 - n_k) n_{k+z} (1 + N_z) \delta(\epsilon_k - \epsilon_{k+z} + \omega_z) + \right.$
 $\left. + (1 - n_k) n_{k-z} N_z \delta(\epsilon_k - \epsilon_{k-z} - \omega_z) - \right.$
 $\left. - n_k (1 - n_{k-z}) (1 + N_z) \delta(\epsilon_{k-z} - \epsilon_k + \omega_z) - \right.$
 $\left. - n_k (1 - n_{k+z}) N_z \delta(\epsilon_{k+z} - \epsilon_k - \omega_z) \right\}$

① $q \rightarrow -q$ $n \rightarrow n$ $k \rightarrow k'$ $+ \textcircled{0}$ $\delta(x-y) = \delta(y-x)$

$$\dot{n}_k = \frac{2\pi}{V} \sum_{k'} |g_\lambda(k, k')|^2 \left\{ (1-n_k) n_{k'} (1+N_2) \delta(\epsilon_k - \epsilon_{k'} + \omega_2) + \right. \\ \left. + (1-n_k) n_{k'} N_2 \delta(\epsilon_k - \epsilon_{k'} - \omega_2) - \right. \\ \left. - n_k (1-n_{k'}) (1+N_2) \delta(\epsilon_k + \epsilon_{k'} - \omega_2) - \right. \\ \left. - n_k (1-n_{k'}) N_2 \delta(\epsilon_k + \epsilon_{k'} + \omega_2) \right\} =$$

$$= \frac{2\pi}{V} \sum_{k'} |g_\lambda(k, k')|^2 \left\{ \delta(\epsilon_k - \epsilon_{k'} - \omega_2) \cdot [(1-n_k) n_{k'} N_2 - (1-n_{k'}) n_k (1+N_2)] + \right. \\ \left. + \delta(\epsilon_k - \epsilon_{k'} + \omega_2) [(1-n_k) n_{k'} (1+N_2) - n_k (1-n_{k'}) N_2] \right\} =$$

Bei Ansatz:

✓ analoges (1.23b)

$$\dot{n}_k = \sum_{k'} \Gamma_{k' \rightarrow k}^{\lambda} (1-n_k) n_{k'} - \Gamma_{k \rightarrow k'}^{\lambda} (1-n_{k'}) n_k$$

$$\Gamma_{k' \rightarrow k}^{\lambda} = \frac{2\pi}{V} |g_\lambda(k, k')|^2 \left\{ (1+N_2) \delta(\epsilon_k - \epsilon_{k'} + \omega_2) + N_2 \delta(\epsilon_k - \epsilon_{k'} - \omega_2) \right\}$$

$$\Gamma_{k \rightarrow k'}^{\lambda} = \frac{2\pi}{V} |g_\lambda(k, k')|^2 \left\{ (1+N_2) \delta(\epsilon_{k'} - \epsilon_k + \omega_2) + N_2 \delta(\epsilon_{k'} - \epsilon_k - \omega_2) \right\}$$

2. BCS currents za ph-el. coupling:

$$k = k_{\uparrow\downarrow} \quad -k = k_{\downarrow\uparrow}$$

$$H_e = \sum_k \xi_k (c_k^\dagger c_k + c_{-k}^\dagger c_{-k}) - \sum_k \left(\Delta c_k^\dagger c_{-k}^\dagger + \Delta^* c_k c_{-k} \right) \quad \begin{matrix} \xi_k = \epsilon_k - \mu \\ \xi_{-k} = \xi_k \end{matrix}$$

$$\rightarrow E_k = \sqrt{\xi_k^2 + \Delta^2}, \quad M_k = \frac{1}{2} \left(1 + \frac{\xi_k}{E_k} \right), \quad N_k = \frac{1}{2} \left(1 - \frac{\xi_k}{E_k} \right), \quad M_k N_k = \frac{\Delta}{2E_k}$$

$$\rightarrow \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} M_k & N_k \\ -N_k & M_k \end{pmatrix} \begin{pmatrix} \gamma_k \\ \gamma_{-k}^\dagger \end{pmatrix} \Rightarrow H_e = E_0 + \sum_{k\sigma} E_k \gamma_{k\sigma}^\dagger \gamma_{k\sigma} \quad N_{k\sigma} = \langle \gamma_{k\sigma}^\dagger \gamma_{k\sigma} \rangle$$

→ enaka interakcija: $H_{ep} = \frac{1}{V} \sum_{\vec{k}, \vec{k}', \sigma} g_{\lambda}(\vec{k}, \vec{k}') c_{\vec{k}+\vec{k}', \sigma}^\dagger c_{\vec{k}, \sigma} (b_{\vec{k}'}^\dagger + b_{-\vec{k}'}^\dagger)$

rotiramo

rotiramo s novo bazo: $(M_k' = M_{k+2}, N_k' = N_k) \quad (M_k' = M_{-k}, N_k' = N_{-k}!)$

$$\begin{aligned} c_{k\uparrow} &= M_k \gamma_{k\uparrow} + N_k \gamma_{-k\downarrow}^\dagger \\ c_{k\uparrow}^\dagger &= M_k \gamma_{k\uparrow}^\dagger + N_k \gamma_{-k\downarrow}^\dagger \end{aligned}$$

$$\begin{aligned} c_{k\downarrow} &= M_k \gamma_{k\downarrow} - N_k \gamma_{-k\uparrow}^\dagger \\ c_{k\downarrow}^\dagger &= M_k \gamma_{k\downarrow}^\dagger - N_k \gamma_{-k\uparrow}^\dagger \end{aligned}$$

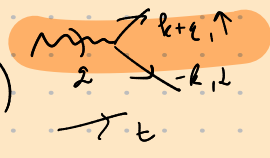
$$\begin{aligned} \downarrow \\ \sum_{k\sigma} c_{k+\vec{q}, \sigma}^\dagger c_{k\sigma} &= \sum_k \left[(M_{k+2} \gamma_{k+2\uparrow}^\dagger + N_{k+2} \gamma_{-k\downarrow}^\dagger) (M_k \gamma_{k\uparrow} + N_k \gamma_{-k\downarrow}^\dagger) + (M_{k+2} \gamma_{k+2\downarrow}^\dagger - N_{k+2} \gamma_{-k\uparrow}^\dagger) (M_k \gamma_{k\downarrow} - N_k \gamma_{-k\uparrow}^\dagger) \right] \\ &= \sum_{k\sigma} M_{k+2} M_k (\gamma_{k+2\uparrow}^\dagger \gamma_{k\sigma}) + \sum_{k\sigma} N_{k+2} N_k (\gamma_{-k\downarrow}^\dagger \gamma_{-k\sigma}) + \\ &+ \sum_k (M_{k+2} N_k \gamma_{k+2\uparrow}^\dagger \gamma_{-k\downarrow}^\dagger - N_{k+2} M_k \gamma_{k+2\downarrow}^\dagger \gamma_{-k\uparrow}^\dagger) \stackrel{h \rightarrow -q-h}{=} (M_{k+2} N_k + M_k N_{k+2}) \gamma_{k+2\uparrow}^\dagger \gamma_{-k\downarrow}^\dagger \\ &+ \sum_k (N_{k+2} M_k \gamma_{-k\downarrow}^\dagger \gamma_{k\uparrow} - M_{k+2} N_k \gamma_{-k\uparrow}^\dagger \gamma_{k\downarrow}) \stackrel{h \rightarrow -q-h}{=} (M_{k+2} N_k + M_k N_{k+2}) \gamma_{-k\downarrow}^\dagger \gamma_{-k\uparrow} \end{aligned}$$

$$\begin{aligned} P_2^+ &= \sum_{k\sigma} c_{k\sigma}^\dagger c_{k\sigma} = 2 \delta_{\vec{q},0} \sum_k N_k^2 + \sum_{k\uparrow\sigma} (M_{k+2} M_k - N_{k+2} N_k) \gamma_{k+2,\sigma}^\dagger \gamma_{k,\sigma} + \\ &+ \sum_k (M_{k+2} N_k + N_{k+2} M_k) [\gamma_{k+2\uparrow}^\dagger \gamma_{-k\downarrow}^\dagger + \gamma_{-k\downarrow}^\dagger \gamma_{k\uparrow}] \end{aligned}$$

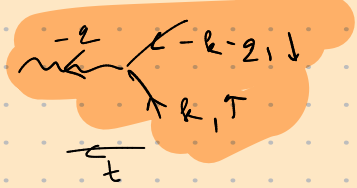
• scattering člena enaka, s faktorjem: $|M_{k+2} M_k - N_{k+2} N_k|^2$

$$\hookrightarrow g_{\lambda}(\vec{k}, \vec{k}') \rightarrow G_{\lambda}^s(\vec{k}, \vec{k}') = g_{\lambda} (M_{k+2} M_k - N_{k+2} N_k)$$

• pair creation: $H_{ep}^c = \frac{1}{V} \sum_{k,q,\lambda} \underbrace{g_{\lambda}(k,q)}_{C_{\lambda}^p(k,q)} (n_{k+q,\uparrow} n_{k,\downarrow} + n_{k+q,\downarrow} n_{k,\uparrow}) \gamma_{k+q,\uparrow}^{\dagger} \gamma_{k,\downarrow}^{\dagger} (b_{2\lambda} + b_{-2\lambda}^{\dagger})$

$$|K_f|H_{ep}^c|i\rangle|^2 \propto \underbrace{|N_{2-1}|b_{2\lambda}|N_{2-1}\rangle|^2}_{N_2} \cdot (1-n_{k+q,\uparrow})(1-n_{k,\downarrow})$$


• pair annihilation: $H_{ep}^a \propto \gamma_{-k+q,\downarrow} \gamma_{k,\uparrow} (b_{2\lambda} + b_{-2\lambda}^{\dagger})$



$$|K_f|H_{ep}^a|i\rangle|^2 \propto \underbrace{|N_{2-1}|b_{2\lambda}^{\dagger}|N_{2-1}\rangle|^2}_{(N_{2-1}+1)} \cdot n_{k,\uparrow} n_{-k+q,\downarrow}$$

celakan tok:

$$\dot{n}_k = \sum_p$$